

Assignment 2 - The LP Model

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Module 2 - Linear Programming Model Formulation

1. Back Savers

Back Savers must decide how many Collegiate and Mini backpacks to produce each week to maximize profit while staying within its nylon, labor, and sales limits.

a. Decision variables

C = number of Collegiate backpacks produced per week

M = number of Mini backpacks produced per week

b. Objective function

Each Collegiate contributes \$32 in profit and each Mini contributes \$24, so total weekly profit is:

$$\text{Maximize } Z = 32C + 24M$$

c. Constraints

Nylon: 5,000 square feet are available; Collegiate uses 3 square feet and Mini uses 2.

$$3C + 2M \leq 5,000$$

Collegiate sales: At most 1,000 per week.

$$C \leq 1,000$$

Mini sales: At most 1,200 per week.

$$M \leq 1,200$$

Labor: 35 laborers x 40 hours x 60 minutes = 84,000 minutes available. Collegiate requires 45 minutes and Mini requires 40 minutes.

$$45C + 40M \leq 84,000$$

Nonnegativity: Production quantities cannot be negative.

$$C, M \geq 0$$

d. Full mathematical formulation

$$\text{Maximize } Z = 32C + 24M$$

subject to

$$3C + 2M \leq 5,000$$

$$C \leq 1,000$$

$$M \leq 1,200$$

$$45C + 40M \leq 84,000$$

$$C, M \geq 0$$

2. Weigelt Corporation

Weigelt Corporation must determine the daily production of large, medium, and small units at each of three plants. The objective is to maximize total daily profit while satisfying plant capacity, storage, sales, and equal-capacity-utilization requirements.

a. Decision variables

Let x_{ij} = number of units of size j produced per day at plant i , where:

- $i = 1, 2, 3$ identifies the plant.
- $j = L, M, S$ identifies the size: large, medium, or small.

Thus, the nine decision variables are $x_{1L}, x_{1M}, x_{1S}, x_{2L}, x_{2M}, x_{2S}, x_{3L}, x_{3M}, x_{3S}$

b. Linear programming model

Objective function. Large, medium, and small units yield profits of \$420, \$360, and \$300, respectively.

$$\text{Maximize } Z = 420(x_{1L} + x_{2L} + x_{3L}) + 360(x_{1M} + x_{2M} + x_{3M}) + 300(x_{1S} + x_{2S} + x_{3S})$$

Plant production-capacity constraints. Plants 1, 2, and 3 can produce at most 750, 900, and 450 units per day, respectively.

$$x_{1L} + x_{1M} + x_{1S} \leq 750$$

$$x_{2L} + x_{2M} + x_{2S} \leq 900$$

$$x_{3L} + x_{3M} + x_{3S} \leq 450$$

In-process storage constraints. Large, medium, and small units require 20, 15, and 12 square feet of storage, respectively.

$$20x_{1L} + 15x_{1M} + 12x_{1S} \leq 13,000$$

$$20x_{2L} + 15x_{2M} + 12x_{2S} \leq 12,000$$

$$20x_{3L} + 15x_{3M} + 12x_{3S} \leq 5,000$$

Sales constraints. Daily sales are limited to 900 large, 1,200 medium, and 750 small units across all three plants.

$$x_{1L} + x_{2L} + x_{3L} \leq 900$$

$$x_{1M} + x_{2M} + x_{3M} \leq 1,200$$

$$x_{1S} + x_{2S} + x_{3S} \leq 750$$

Equal percentage of excess capacity used. Management requires each plant to use the same percentage of its available excess production capacity. Writing the proportional requirement in linear equality form gives:

$$900(x_{1L} + x_{1M} + x_{1S}) - 750(x_{2L} + x_{2M} + x_{2S}) = 0$$

$$450(x_{2L} + x_{2M} + x_{2S}) - 900(x_{3L} + x_{3M} + x_{3S}) = 0$$

Nonnegativity. All production quantities must be nonnegative.

$$x_{ij} \geq 0 \quad \text{for } i = 1, 2, 3 \text{ and } j = L, M, S$$

Full mathematical formulation - Weigelt Corporation

$$\text{Maximize } Z = 420(x_{1L} + x_{2L} + x_{3L}) + 360(x_{1M} + x_{2M} + x_{3M}) + 300(x_{1S} + x_{2S} + x_{3S})$$

subject to

$$x_{1L} + x_{1M} + x_{1S} \leq 750$$

$$x_{2L} + x_{2M} + x_{2S} \leq 900$$

$$x_{3L} + x_{3M} + x_{3S} \leq 450$$

$$20x_{1L} + 15x_{1M} + 12x_{1S} \leq 13,000$$

$$20x_{2L} + 15x_{2M} + 12x_{2S} \leq 12,000$$

$$20x_{3L} + 15x_{3M} + 12x_{3S} \leq 5,000$$

$$x_{1L} + x_{2L} + x_{3L} \leq 900$$

$$x_{1M} + x_{2M} + x_{3M} \leq 1,200$$

$$x_{1S} + x_{2S} + x_{3S} \leq 750$$

$$900(x_{1L} + x_{1M} + x_{1S}) - 750(x_{2L} + x_{2M} + x_{2S}) = 0$$

$$450(x_{2L} + x_{2M} + x_{2S}) - 900(x_{3L} + x_{3M} + x_{3S}) = 0$$

$$x_{ij} \geq 0 \quad \text{for } i = 1, 2, 3 \text{ and } j = L, M, S$$

Modeling note: The equal-utilization requirement is modeled with linear equality constraints, following the Chapter 3 approach for situations in which multiple locations must use the same proportion of available capacity. The objective and all constraints remain linear.